Semi-Synthetic Training with Empirically Learned Noise for ppm-Level H_2S Retrieval in Open-Path UV-DOAS

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***Abstract*—Quantitative retrieval of hydrogen sulfide (H_2S) using ultraviolet Differential Optical Absorption Spectroscopy (UV-DOAS) under realistic open-path conditions is strongly affected by structured, wavelength-dependent instrumental and environmental perturbations that are poorly represented by simplified noise models. This work proposes a semi-synthetic training strategy combining a Beer–Lambert forward model with structured noise learned from experimental UV-DOAS residuals using an autoencoder–Gaussian mixture model (AE–GMM). The resulting dataset provides continuous concentration labels while preserving experimentally observed spectral variability. The retained CNN estimator was evaluated on 886 experimental spectra acquired during three campaigns at nominal source–detector distances of 15, 45, and 80 m, without distance-specific calibration. The method achieved a global MAE of 13.43 ppm and an RMSE of 17.01 ppm, with a mean signed error of +0.30 ppm. No monotonic degradation of retrieval performance with path length was observed across the three campaigns. These results demonstrate the effectiveness of experimentally grounded semi-synthetic training for ppm-level H_2S retrieval from open-path UV-DOAS measurements.**

***Index Terms*—Gas sensing, Measurement uncertainty, Semi-synthetic data, Sparse experimental data, Spectroscopy**

I. Introduction

The real-time detection and monitoring of hydrogen sulfide H_2S emissions are essential due to its extreme toxicity [1] and its widespread presence across various industrial sectors. Being both corrosive and hazardous, H_2S presents major challenges in industries such as oil and gas exploration [2], refining, and chemical manufacturing [3]. In the oil and gas industry, where it is commonly found in natural gas and crude oil, it poses significant risks to worker safety and infrastructure integrity [4]. Similarly, in wastewater treatment plants [5] and sewer networks [6], H_2S generated by the decomposition of organic matter can cause severe equipment corrosion and serious health hazards. Accurate monitoring and reliable quantification of H_2S concentration are therefore critical requirements for maintaining safety and operational efficiency in industrial environments.

However, existing real-time detection technologies remain limited under realistic operating conditions. Conventional analytical methods, such as gas chromatography or lead acetate sensing tapes, typically require sample transportation to dedicated laboratories and do not provide continuous monitoring capabilities. In addition, the high reactivity of H_2S may introduce significant biases during sampling and handling, leading to inaccuracies in concentration estimation and limiting their applicability in dynamic environments.

Optical absorption spectroscopy techniques have been investigated for gas sensing applications, including Tunable Diode Laser Absorption Spectroscopy (TDLAS) [7] and Fourier Transform Infrared Spectroscopy (FTIR). While TDLAS offers high sensitivity and selectivity by targeting specific absorption lines [8], and FTIR provides broadband detection capabilities, their deployment in outdoor or industrial environments remains challenging. Instrumental complexity, calibration drift, and sensitivity to environmental perturbations often limit their long-term robustness under real operating conditions.

In this work, we focus on ultraviolet differential optical absorption spectroscopy (UV-DOAS) [9] operating in the 203–$286\,\mathrm{nm}$ spectral range, where H_2S exhibits strong and distinctive absorption features. This wavelength interval is fully covered by a broadband xenon lamp, enabling continuous spectral illumination suitable for open-path measurements. Under controlled conditions, the absorption process is accurately described by the Beer–Lambert law and does not constitute the primary limitation for concentration retrieval.

In outdoor UV-DOAS configurations, a major limitation arises instead from complex, wavelength-dependent spectral perturbations. These perturbations result from the combined effects of atmospheric scattering (Rayleigh and Mie), broadband extinction, source intensity fluctuations, detector spectral response, and electronic noise [10], [11]. These contributions are spectrally non-uniform and statistically correlated, with dependencies on both the instrument configuration and the measurement environment [12], [13]. Simplified independent noise models may therefore be insufficient and empirical characterization from experimental data is required.

Two fundamental limitations constrain the generalization of data-driven approaches to realistic outdoor gas sensing. First, experimentally labeled datasets covering wide concentration ranges and realistic environmental conditions remain scarce:

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extensive calibration campaigns are both costly and difficult to conduct in outdoor settings. Second, measurement variability is strongly influenced by instrument- and environment-specific spectral perturbations, which exhibit wavelength dependence and structured correlations. These effects are poorly reproduced by simplified synthetic noise models, which can limit model transfer across instruments, propagation distances, or environmental conditions.

Data-driven methods have been increasingly used for spectroscopic gas sensing and concentration retrieval. Classical machine-learning approaches, such as support-vector-machine-based methods, have been applied to gas classification and quantification tasks [14]. More recent studies have used deep-learning models to extract concentration information from high-dimensional spectral measurements, including rotational, infrared, tunable-diode-laser, and optical absorption spectra [15]–[19]. Data-driven strategies have also been proposed for H_2S quantification using virtual-sensing approaches coupled with physical gas sensors [20].

However, these approaches do not directly address the specific calibration problem considered here: constructing training data that remain physically consistent while reproducing the structured perturbations observed in outdoor UV-DOAS (H_2S) measurements. This gap motivates the semi-synthetic strategy developed in this work.

Preliminary investigations during the methodological development of the retrieval approach indicated that direct training on a limited number of experimental spectra provided restricted transferability when acquisition conditions differed from those represented in the training data. In such datasets, concentration information may remain partially entangled with campaign-specific spectral variability, allowing the regression model to learn acquisition-dependent signatures rather than only the physical concentration dependence of the absorption signal. Conventional Gaussian augmentation increases data diversity but does not adequately reproduce the wavelength-dependent and correlated structure observed in experimental UV–DOAS residuals. These observations motivated the semi-synthetic strategy adopted in the present work, in which concentration-dependent absorption is generated independently from the Beer–Lambert model over the full concentration range, while structured spectral perturbations are learned separately from experimental residuals and subsequently combined with the physical spectra. This formulation also enables the construction of a large and densely sampled training dataset that cannot be obtained from the limited number of experimentally labeled spectra alone, while exposing the regression model to a broad range of combinations between concentration and experimentally representative spectral variability.

The available experimental dataset was acquired over multiple campaigns and under varying propagation distances and environmental conditions. Although it is not sufficient by itself for direct supervised CNN training, it provides the experimental basis for learning realistic spectral perturbations. Physically consistent absorption spectra are generated using the Beer–Lambert law and reference cross-sections from the MPI–Mainz UV/VIS Spectral Atlas [21]. Experimentally derived structured noise is then injected to construct the semi-synthetic training datasets. The complex distribution of these perturbations is represented through an autoencoder latent space modeled by a Gaussian mixture model (GMM).

The main contribution of this work is a measurement-oriented semi-synthetic calibration strategy that separates the physical concentration-dependent absorption model from experimentally learned spectral variability. The retained estimator was evaluated on 886 experimental spectra acquired during three campaigns at nominal source–detector distances of 15, 45, and 80 m, achieving a global MAE of 13.43 ppm and an RMSE of 17.01 ppm, with a mean signed error of +0.30 ppm. These results support the importance of realistic noise representation in the construction of semi-synthetic training data for open-path UV-DOAS concentration retrieval.

The paper follows a progressive structure, from experimental grounding to data-driven validation. Section II describes the experimental setup and the UV-DOAS measurement configuration. Section III introduces the forward physical model used for the generation of physically consistent absorption spectra. Section IV details the empirical noise characterization and the data generation pipeline. Section V describes the semi-synthetic dataset construction and the regression framework. Section VI presents the experimental validation and comparative concentration retrieval performance. Section VII discusses the generalization domain and current limitations of the approach. Finally, Section VIII summarizes the main findings.

II. Experimental Setup Description

The experimental setup serves two complementary objectives. First, it provides controlled absorption measurements for checking whether the acquired spectra follow Beer–Lambert predictions, used here as a first-order approximation of the true and inherently intractable physical process. Second, and of primary relevance to the present study, it enables the acquisition of realistic spectral noise profiles arising from the combined effects of optical instrumentation, gas management, and environmental variability. These experimentally characterized noise contributions form the empirical foundation of the data-driven noise model introduced in Section IV.

A schematic representation of the experimental system is shown in Figure 1. The light source is a Xenon flash lamp (Hamamatsu L6565) delivering a stable broadband emission spanning the deep ultraviolet to the near-infrared, with usable power across the 180–$400\,\mathrm{nm}$ range and a peak optical power of up to $10\,\mathrm{W}$. The short-term output fluctuations are below 0.005%, and the long-term stability drift remains within 0.3% per hour. A band-pass filter spanning 100–$400\,\mathrm{nm}$ is placed immediately after the source to suppress higher-order diffraction and stray-light contributions originating from longer wavelengths, given the grating-based architecture of the spectrometer and the inherently broadband nature of the source.

The transmitted light is recorded using a UV–VIS spectrometer (OTO Photonics, model EE2063), covering the spectral range 180–$1100\,\mathrm{nm}$. The instrument provides a nominal

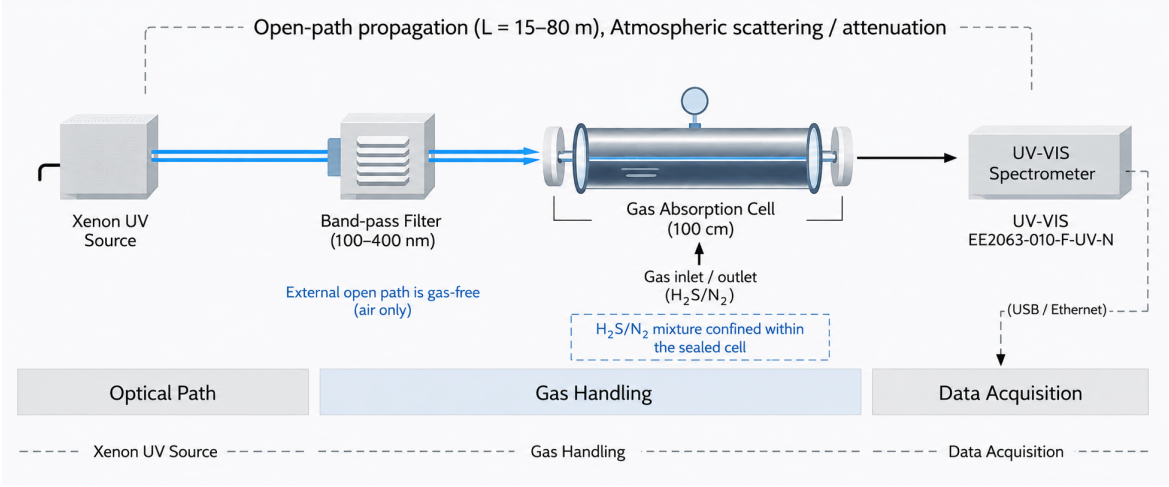


Fig. 1: Schematic representation of the UV-DOAS experimental setup. Broadband ultraviolet radiation emitted by a Xenon source propagates over an open atmospheric path of variable length L where wavelength-dependent scattering and attenuation occur. The beam is spectrally filtered and subsequently transmitted through a sealed gas absorption cell of $100\,\mathrm{cm}$ optical length. The cell is filled with calibrated H_2S/N_2 mixtures via dedicated inlet and outlet ports, while the external optical path remains gas-free. The transmitted spectrum is recorded by a UV–VIS spectrometer and transferred to the data acquisition system for storage and analysis.

optical resolution of $1.7\,\mathrm{nm}$ with a slit width of $10\,\mu\mathrm{m}$, and exhibits a stray-light level of approximately 0.45% as specified by the manufacturer. Spectra are acquired at varying H_2S concentrations and stored on a fixed instrumental wavelength grid.

The gas absorption cell consists of a stainless-steel tube with an effective optical path length of $100\,\mathrm{cm}$, sealed at both ends with sapphire windows. The internal optical path includes a single reflection, and all internal surfaces are Teflon-coated to reduce adsorption effects associated with H_2S. The cell is filled with calibrated mixtures of H_2S diluted in high-purity nitrogen over a concentration range spanning 0–$800\,\mathrm{ppm}$. Gas mixtures are produced using a custom-built gas blender capable of dilution ratios up to $300:1$, enabling coverage of a wide concentration range with a distribution accuracy of $\pm 1\%$ at full scale. Key characteristics of the Xenon source and the gas cell are summarized in Table I.

Broadband Xenon Source	
Wavelength range	$180\,\mathrm{nm} - 1700\,\mathrm{nm}$
Optical power	$10\,\mathrm{W}$
Intensity distortion	0.05%
Repetition rate	$1\,\mathrm{Hz}$
Beam diameter (collimated)	$6\,\mathrm{mm}$
Gas Cell	
Equivalent optical length	$100\,\mathrm{cm}$
Window material	Sapphire
Gas mixture	H_2S / N_2

TABLE I: Main characteristics of the broadband Xenon source and the gas absorption cell.

At each concentration level and propagation distance, several hundred spectra are acquired over extended sessions spanning multiple hours, different times of day, and different seasons, ensuring broad coverage of instrumental and environmental variability. This strategy provides a statistically rich basis for both model validation and noise characterization, as detailed in Section IV.

Measurements were conducted under semi-controlled conditions: gas concentration, optical path length, alignment, and acquisition parameters were strictly controlled, while environmental factors such as ambient temperature, residual humidity, and long-term source drift were intentionally left unstabilized. Outdoor acquisitions were performed in a sheltered tunnel environment, providing natural shading and wind protection. Ambient temperature remained within 25–$35\,^\circ\mathrm{C}$, consistent with the conditions under which the reference absorption cross-sections are tabulated [21]. Propagation-induced effects, including atmospheric backscattering and signal attenuation, were partially compensated through dedicated amplification and signal conditioning electronics. This approach allows wavelength-dependent perturbations to emerge naturally in the recorded spectra, yielding a physically representative dataset for the noise model developed in Section IV.

All spectra are projected onto a common instrumental wavelength grid of 192 uniformly spaced points over the 197–$292\,\mathrm{nm}$ range. Since the effective optical resolution of the spectrometer is significantly broader than the native spectral resolution of the MPI–Mainz reference cross-sections [21], the latter are resampled at $0.2\,\mathrm{nm}$ and convolved with the instrumental line shape, approximated by a Gaussian profile consistent with the grating-based architecture of the spectrometer, prior to their use in the forward model. This procedure ensures spectral consistency between the reference cross-sections and the experimentally acquired spectra.

By adjusting the gas mixture concentration, a set of labeled experimental spectra is acquired under controlled conditions. Figure 2 presents an overlay of representative acquisitions, including the dark spectrum, the reference spectrum obtained with pure N_2, and measurements at selected H_2S

concentrations. Beyond the expected concentration-dependent absorption features, the overlay reveals baseline distortions and wavelength-dependent intensity variations that cannot be attributed to molecular absorption alone, and which constitute the primary motivation for the noise characterization framework developed in Section IV.

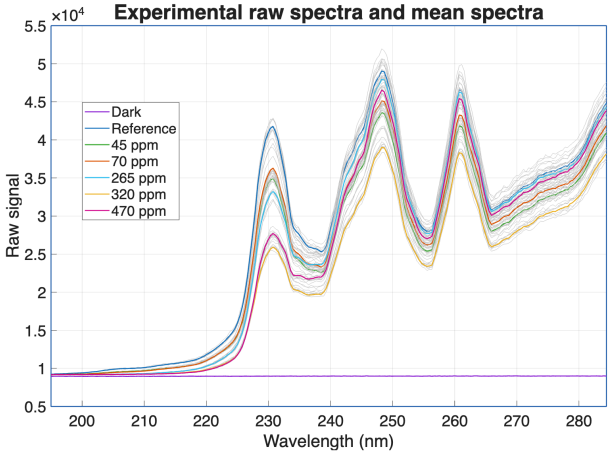


Fig. 2: Experimental UV-DOAS spectra acquired on the native wavelength grid for the dark signal, the N_2 reference, and five selected H_2S concentrations. Thin curves show representative acquisitions, while bold curves indicate the corresponding mean spectra. The overlay highlights the raw spectral variability and the concentration-dependent evolution of the measured signal before normalization.

However, these raw signals are expressed in arbitrary instrumental units reflecting detector gain, integration time, and spectral response, and cannot be directly interpreted within a physical absorption framework. To address this, the measured intensity is transformed into a normalized transmission signal following Eq. 3, as defined in Section III. The resulting spectra, shown in Fig. 3, remove the instrumental offset and yield a physically consistent representation of the absorption behavior across all experimental conditions.

III. Forward Absorption Model

This section defines the physical model linking gas concentration to the measured spectral observable, without performing any analytical inversion.

A. Cell–atmosphere propagation model

In the present configuration, combining a sealed gas cell of optical path length L_c and an open atmospheric path of total length L, the Beer–Lambert law gives the detected intensity as

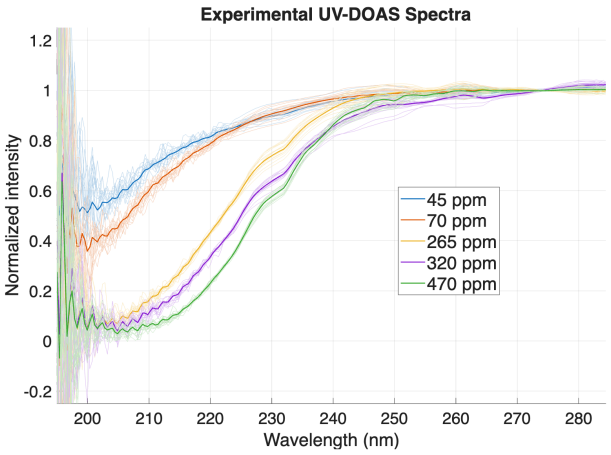


Fig. 3: Normalized transmission spectra after quality filtering at a propagation distance of 45 m. Thin curves show selected experimental acquisitions, while thick curves indicate the corresponding mean spectra.

$$I_f(\lambda) = I_0(\lambda) \times \exp\left[-\int_0^{L_c}\Big(\sum_{k\in\{\mathrm{H_2S,N_2}\}}\sigma_k(\lambda,T)c_k(x)+\alpha_c(\lambda,x)\Big)\,dx\right] \times \exp\left[-\int_{L_c}^{L}\Big(\sum_j \sigma_j(\lambda,T)c_j(x)+\alpha_a(\lambda,x)\Big)\,dx\right], \tag{1}$$

where $\sigma_k(\lambda,T)$ and $c_k(x)$ denote the absorption cross-section and local concentration of species k, and $\alpha_c(\lambda,x)$, $\alpha_a(\lambda,x)$ represent effective extinction contributions within the cell and along the atmospheric path, respectively, encompassing Rayleigh and Mie scattering and transmission losses.

Equation (1) accounts for a heterogeneous medium with multiple absorbing species and spatially varying concentrations. In practice, the analysis is restricted to narrow spectral windows centered on specific absorption features of the target gas, within which the measured transmission deviates from the ideal Beer–Lambert formulation through additive and multiplicative spectrally structured perturbations — collectively referred to as the spectral baseline — arising from instrumental response, atmospheric variability, and propagation effects. These perturbations are not explicitly modeled within the forward framework but are addressed through the semi-synthetic noise generation strategy described in Section IV.

In the present configuration, H_2S is confined to the sealed absorption cell of length L_c, while the atmospheric path remains gas-free. Outdoor propagation introduces wavelength-dependent perturbations — including atmospheric scattering, turbulence-induced fluctuations, and distance-dependent attenuation — whose cumulative influence is embedded in the experimental spectral residuals acquired over multiple campaigns, from which the structured noise representation is learned as described in Section IV.

Within the sealed cell, the spatial uniformity of H_2S concentration allows the path integral to be evaluated exactly, yielding

$$I_f(\lambda) = I_0(\lambda) \exp[-\sigma(\lambda)\, C\, L_c]\,, \tag{2}$$

where $C \cdot L_c$ is the column density of the absorbing species within the cell, and $\sigma(\lambda)$ denotes the H_2S absorption cross-section preprocessed as described in Section II.

B. Definition of the observed normalized intensity

In practical measurements, the idealized Beer–Lambert formulation of Eq. (2) is affected by instrumental offsets and acquisition artifacts that prevent the direct use of raw intensity ratios. The measured spectra are therefore corrected for the dark signal and normalized with respect to a reference acquisition, defining the spectral observable used throughout this study.

The normalized experimental spectrum $\tilde{I}_{\mathrm{e}}(\lambda)$ is defined as

$$\tilde{I}_{\mathrm{e}}(\lambda) = \frac{I_{f,\mathrm{ppm}}(\lambda) - I_{f,\mathrm{dark}}(\lambda)}{I_{f,0}(\lambda) - I_{f,\mathrm{dark}}(\lambda)}, \tag{3}$$

where $I_{f,0}(\lambda)$ is the reference spectrum acquired with pure N_2, $I_{f,\mathrm{ppm}}(\lambda)$ the spectrum transmitted through the gas-containing medium, and $I_{f,\mathrm{dark}}(\lambda)$ the dark spectrum. This normalization removes the instrumental background and establishes the physically consistent observable upon which the semi-synthetic perturbation framework is constructed.

C. Training conditions and comparative framework

To isolate the contribution of noise modeling fidelity to concentration retrieval performance, three training conditions are systematically compared, all evaluated on the same independent experimental test data. In the first condition, the regression model is trained on noise-free spectra generated directly from Eq. (2), providing a baseline that quantifies performance in the absence of any noise representation. In the second condition, independent Gaussian noise is superimposed on the synthetic spectra, representing the standard augmentation strategy in the literature. In the third and proposed condition, empirically learned structured noise extracted from long-term experimental observations is injected into the synthetic spectra, as detailed in Section IV. This progressive framework directly attributes performance differences to the realism of the noise representation, independently of the regression architecture.

IV. Experimental Noise Characterization and Data-Driven Generation

The noise affecting the measured transmission spectra is characterized directly from experimental field data in order to preserve realistic measurement conditions. Rather than reproducing individual acquisitions, the proposed approach focuses on extracting the statistical structure of experimental perturbations for controlled noise generation. These structured perturbations arise from a combination of instrumental response, atmospheric fluctuations, and propagation effects, leading to wavelength-dependent distortions that cannot be modeled as independent noise.

A. Residual extraction and filtering

For each concentration level c, residual noise is defined as the deviation of an individual spectrum from the empirical mean spectrum computed over all acquisitions at the same concentration:

$$N_i^{(c)}(\lambda) = S_i^{(c)}(\lambda) - \langle S^{(c)}(\lambda) \rangle, \tag{4}$$

where $S_i^{(c)}(\lambda)$ denotes the i-th measured spectrum and $\langle S^{(c)}(\lambda) \rangle$ its empirical mean. This operation removes the dominant deterministic absorption component and isolates stochastic perturbations arising from instrumental response, atmospheric variability, and measurement fluctuations.

Residuals are subsequently filtered using a wavelength-resolved median absolute deviation (MAD) criterion to ensure consistency across acquisition sets collected at different propagation distances and amplification stages. A robust threshold of three local MADs is applied at each wavelength, resulting in the rejection of approximately 2–3% of residual samples. This limited rejection rate indicates that the filtering primarily removes isolated transient artifacts or abnormal amplifier responses, while preserving the statistical structure of the experimental noise corpus used in the subsequent generative stage.

B. Data-driven structured noise generation

Experimental noise exhibits wavelength-dependent structures that cannot be reproduced by simple uncorrelated perturbations. To capture these effects, a data-driven approach is adopted, where noise patterns are learned directly from experimental residuals and used to generate realistic perturbations for semi-synthetic training.

1) Autoencoder-based structured noise modeling: The structured component of the experimental noise is modeled using a fully connected autoencoder trained on residual spectra. The objective is to learn a compact representation of wavelength-dependent perturbations while preserving their dominant spectral organization.

The network follows a symmetric encoder–decoder architecture. Starting from the 192-dimensional input residual vector, the encoder progressively compresses the signal through two hidden layers of dimensions 128 and 64 down to a latent space of dimension 24. The decoder mirrors this structure through layers of dimensions 64 and 128 and reconstructs the signal in the original spectral space. Rectified linear unit (ReLU) activations are used in the hidden layers, while the output layer remains linear to preserve amplitude variations. The detailed architecture is summarized in Table II. A fully connected architecture was adopted because the residual spectra are represented by a moderate-dimensional vector after wavelength selection and resampling. This architecture provides sufficient capacity to capture broadband wavelength correlations without introducing convolutional assumptions that are not required at the considered spectral resolution.

Latent dimensions in the range 16–48 were evaluated on a held-out validation set. A latent dimension of 24 was retained

TABLE II: Architecture of the autoencoder used for structured noise modeling

Layer	Type	Output size	Activation
Input	Feature input	192	–
FC1	Fully connected	128	ReLU
FC2	Fully connected	64	ReLU
Latent	Fully connected	24	Linear
FC3	Fully connected	64	ReLU
FC4	Fully connected	128	ReLU
Output	Fully connected	192	Linear

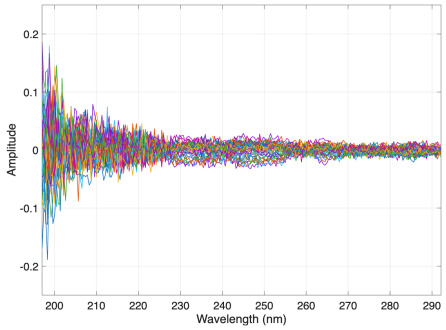
because it provided a low reconstruction error while maintaining a compact representation of the dominant wavelength-dependent residual structure. Increasing the latent dimension beyond this value did not lead to a significant improvement in reconstruction performance.

The latent representations exhibit non-Gaussian statistics, including multimodality and heavy-tailed behavior, indicating the presence of multiple experimental noise regimes.

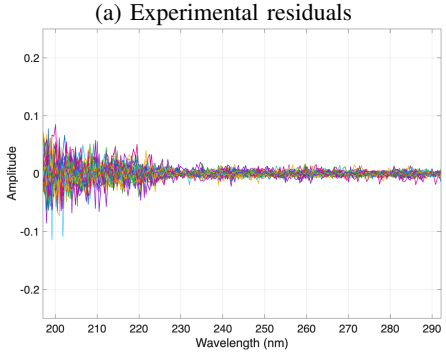
Analysis of the latent representations reveals non-Gaussian behavior, including multimodal distributions and heavy tails, reflecting the presence of multiple noise regimes associated with varying measurement conditions such as changes in signal level, atmospheric stability, and detector response. As demonstrated in Section VI (Fig. 7), direct Gaussian latent sampling leads to a marked compression of the generated RMS distribution relative to the experimental reference,confirming that Gaussian sampling is insufficient to reproduce the full statistical variability of experimental noise. A Gaussian Mixture Model (GMM) is therefore fitted to the latent space to capture this higher-order statistical structure. The number of mixture components is selected using the Bayesian Information Criterion (BIC), as detailed in Section VI.

Representative examples of experimental residuals, autoencoder reconstructions, and generated noise realizations are shown in Fig. 4. The purpose of this figure is mainly illustrative: it shows that the autoencoder does not only reproduce pointwise amplitude variations, but also keeps part of the broadband and wavelength-correlated structure observed in the residual spectra.

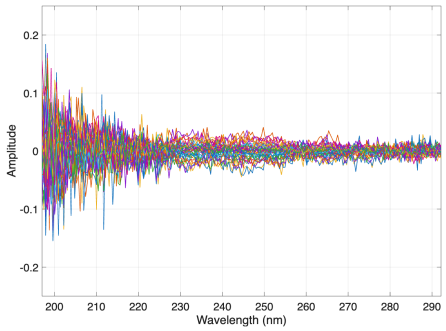
New noise realizations are generated by sampling the GMM-fitted latent distribution and decoding the sampled latent vectors through the trained autoencoder decoder. Each realization, denoted $N_{\mathrm{struct}}(\lambda)$, is then centered and rescaled according to the experimental RMS statistics of the residuals $N_i^{(c)}(\lambda)$ defined in Eq. (4). These realizations are used as the structured noise component in the semi-synthetic dataset described in Section V.



(a) Experimental residuals



(b) Autoencoder reconstruction



(c) Generated noise

Fig. 4: Structured noise modeling for semi-synthetic data generation: (a) experimental residuals, (b) autoencoder reconstruction, and (c) generated noise sampled from the learned latent distribution.

C. Experimental Noise Dataset and Statistical Modeling

The experimental noise corpus is constructed from residual spectra extracted from outdoor measurements acquired under different propagation distances, amplification settings, and environmental conditions. After concentration-wise mean subtraction and MAD-based filtering, the residuals form a heterogeneous but statistically controlled corpus over the 197–292,nm spectral range, sampled at 192 uniformly spaced wavelength points.

Propagation distance is not introduced as an explicit parameter in the generative model. Its effect is instead embedded in the residual corpus through the variations of signal level, relative noise amplitude, and spectral distortions observed in the measurements. This choice avoids imposing a parametric distance-dependent noise law, while still allowing the learned noise distribution to reflect the range of experimental acquisition conditions.

The filtered residual corpus is used to train the autoencoder–GMM model described above. Once trained, this model provides a compact statistical representation of experimental perturbations and allows large ensembles of structured noise realizations to be generated efficiently by latent-space sampling and decoding. These realizations, denoted $N_{\mathrm{struct}}(\lambda)$, are then used as the noise component injected into the semi-synthetic spectra described in Section V.

V. Semi-Synthetic Dataset Construction

This section describes the generation of the semi-synthetic spectra used for supervised CNN training. Following recent work showing that deep neural networks can retrieve gas concentrations from optical spectra [16], the proposed approach uses a CNN regression model; however, its training set is generated from a Beer–Lambert forward model combined with experimentally learned noise perturbations, rather than from experimental concentration-labelled spectra.

Experimental spectra are therefore not used for CNN training. They are used only for noise characterization and for independent validation under real operating conditions.

A. Physically grounded semi-synthetic spectrum generation

The limited availability of fully labeled field measurements is a major limitation for supervised learning in open-path UV absorption spectroscopy. In outdoor conditions, acquiring large datasets over a wide range of concentrations, path lengths, and atmospheric states is costly and difficult to control. A semi-synthetic strategy is therefore used to generate labeled training spectra while keeping the forward physical model explicit.

Noise-free theoretical spectra are generated from the reduced sealed-cell forward model of Eq. (2), using the H_2S absorption cross-section $\sigma(\lambda)$ from the MPI–Mainz UV/VIS Spectral Atlas [21]. They are computed on the experimental wavelength grid, providing ideal transmission spectra $I_{\mathrm{f}}(\lambda)$ with known concentration labels.

A structured noise realization $N_{\mathrm{struct}}(\lambda)$, generated from the autoencoder–GMM noise model described in Section IV, is then added in the measurement domain:

$$I_{\mathrm{syn}}(\lambda) = I_{\mathrm{f}}(\lambda) + N_{\mathrm{struct}}(\lambda). \tag{5}$$

For each generated event, the concentration is sampled continuously within the prescribed concentration range and is used as the exact label of the Beer–Lambert forward calculation. The structured perturbation is sampled independently from the experimentally learned noise distribution. The resulting dataset therefore combines continuous concentration labels with approximately uniform coverage of the prescribed range and experimentally grounded spectral variability.

The semi-synthetic dataset is not intended as a substitute for independent experimental validation. Its role is to extend the labeled training space under noise conditions derived from real measurements. The overall interaction between physical modeling, experimental noise generation, and concentration retrieval is summarized in Fig. 5.

B. Training dataset and preprocessing

For the experiments reported in this work, the semi-synthetic training dataset was generated over the concentration range from 0 to 750 ppm using continuous stratified random sampling. The concentration domain was divided into 80 equal-width intervals, from which a total of 100,000 concentration values were sampled, providing nearly uniform coverage of the complete range without repeatedly using a limited set of fixed labels. Each concentration value was combined with an independently generated structured-noise realization. The concentration label was therefore defined by the Beer–Lambert forward model, while the spectral perturbation was sampled from the experimentally learned noise distribution.

The original instrumental spectra contained 192 physical wavelength channels over the selected UV range. Before CNN processing, each spectrum was interpolated onto a fixed grid of 960 positions. This operation increased the numerical sampling density available to the convolutional layers without introducing additional physical spectral information.

Semi-synthetic and experimental spectra were processed using the same preprocessing chain. Wavelet denoising was performed using a Symlet 4 decomposition with Block James–Stein thresholding. Each spectrum was then represented by two wavelength-dependent channels: the normalized spectral response and its first derivative with respect to wavelength. The first channel retained the overall absorption depth and spectral shape, whereas the derivative channel emphasized local spectral variations.

Normalization parameters were estimated exclusively from the semi-synthetic training subset and subsequently applied unchanged to the internal validation subset and to all experimental measurements. A held-out semi-synthetic subset was used only for training monitoring and early stopping. Experimental measurements were completely excluded from model fitting and hyperparameter selection and were introduced only after training for independent validation under real operating conditions.

C. CNN regression model

The concentration estimator is implemented as a one-dimensional CNN operating along the wavelength dimension. The network receives a two-channel spectral input and returns a continuous scalar output corresponding to the estimated H_2S concentration in ppm. In this work, the CNN is used as a regression estimator to evaluate the transfer of the semi-synthetic training strategy to independent experimental measurements, rather than being introduced as a new neural-network architecture.

The feature-extraction stage comprises four convolutional blocks with 64, 128, 256, and 512 filters and kernel sizes of 7, 5, 3, and 3, respectively. Each convolutional layer is followed by batch normalization and a leaky-ReLU activation. Max pooling is applied after the third convolutional block, and dropout is introduced after the final block.

Each input spectrum is represented by two complementary wavelength-dependent channels. The first channel contains the normalized spectral response and retains the absorption

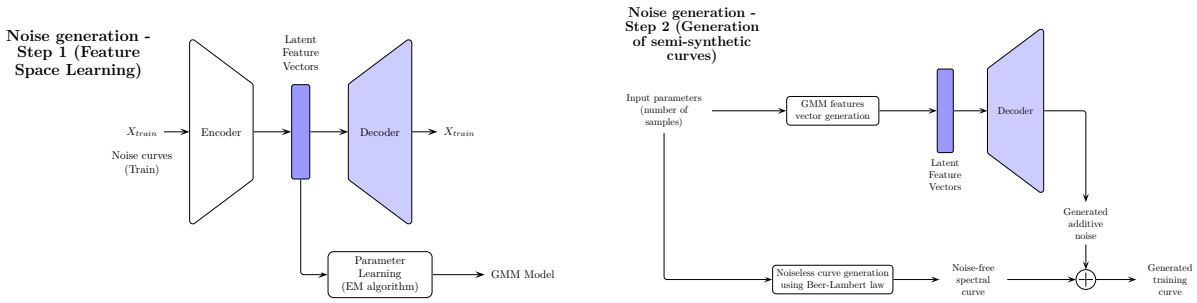


Fig. 5: Overview of the semi-synthetic measurement pipeline for H_2S concentration retrieval. Noise-free absorption spectra are generated using the Beer–Lambert forward model. Empirically learned structured noise is then injected to produce semi-synthetic spectra with exact concentration labels. The CNN is trained on this dataset and applied unchanged to experimental measurements acquired under real operating conditions.

amplitude, baseline behavior, and spectral shape. The second channel contains the first wavelength derivative of the same processed spectrum and describes its local spectral slope. Positive and negative values identify rising and falling spectral regions, while the derivative magnitude represents the local rate of variation. This derived channel emphasizes the positions and shapes of spectral transitions that may be less apparent from the normalized intensity alone; it does not constitute an additional measurement.

The final convolutional feature maps are converted into a feature vector using a Flatten layer and subsequently processed by three fully connected layers containing 256, 128, and 64 neurons. In contrast to global averaging, this operation allows the dense regression layers to retain access to wavelength-position-specific activations extracted from both input channels. The final linear layer provides the concentration estimate.

Training is performed using the Adam optimizer with a mean squared error loss, a mini-batch size of 128, and an initial learning rate of 10^{-3}. L2 regularization, dropout, and early stopping based on the held-out semi-synthetic validation loss are used to limit overfitting. The network corresponding to the lowest validation loss is retained for experimental evaluation.

No source–detector distance, propagation length, light-source identifier, electronic gain, or hardware-configuration parameter is supplied to the network. Concentration retrieval therefore relies exclusively on the information contained in the two-channel spectral input.

Several regression configurations were examined during model development, including alternative feature-aggregation strategies and input representations. The retained configuration combines the processed spectrum with its wavelength derivative as a two-channel input and uses a Flatten layer to provide the dense regression stages with wavelength-position-specific convolutional activations. This configuration was selected exclusively from the semi-synthetic training and validation data because it provided the most stable convergence and validation behavior. A systematic ablation of the input representation, preprocessing strategy, feature-aggregation method, and CNN architecture is beyond the scope of the present experimental-transfer study.

VI. Results

This section evaluates the proposed semi-synthetic training strategy on experimental measurements not included in the semi-synthetic training or validation datasets. The CNN estimator was trained using spectra generated from the reduced Beer–Lambert forward model and the AE–GMM residual-noise model. The experimental spectra were not used for network-weight optimization or early stopping.

The results reported below were obtained using the retained two-channel Conv–Flatten regression configuration described in Section **??**. The reported performance should therefore be interpreted as that of the complete estimation pipeline, including spectral preprocessing, the two-channel input representation, and the CNN regressor, rather than as an architecture-independent property of the semi-synthetic dataset alone.

The results are organized as follows. First, the consistency of the learned noise model is assessed by comparing experimental residuals with generated perturbations. Second, the retained estimation pipeline is evaluated on experimental spectra using global performance metrics and prediction–reference agreement. Third, retrieval performance is compared across measurement campaigns conducted at different nominal path lengths and under campaign-specific acquisition conditions. Finally, the error behavior is examined as a function of reference concentration.

A. Validation of the Learned Noise Model

Before evaluating concentration retrieval, the AE–GMM noise generator was examined by comparing generated perturbations with experimental residuals. This analysis verifies whether the perturbations used for semi-synthetic training reproduce the main amplitude characteristics observed in the experimental measurements.

1) Residual Statistics Versus Concentration: The residual spectra defined in Eq. (4) were analyzed as a function of gas concentration to determine whether their amplitude was dominated by a concentration-dependent behavior. For each concentration level, the per-spectrum root-mean-square (RMS) residual amplitude was computed over the complete spectral interval after concentration-wise mean subtraction and appli-

cation of the residual-processing procedure described in the methodology.

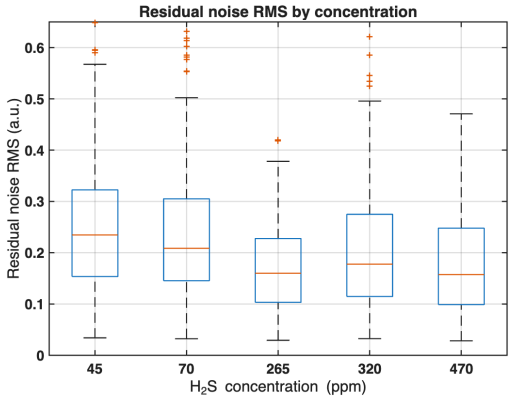


Fig. 6: RMS amplitude of the experimental residuals grouped by H_2S concentration for the measurement campaign conducted at a nominal source–detector path length of 45 m.

Fig. 6 presents the residual RMS distributions obtained from the 45-m campaign, which provided a sufficiently large set of repeated measurements acquired under stable settings within the campaign. Although variations are observed between concentration groups, the distributions remain broadly overlapping over the investigated range. The residual amplitude therefore does not exhibit a dominant monotonic dependence on concentration.

This result supports the separation adopted in the semi-synthetic generation procedure, in which the Beer–Lambert forward model imposes the concentration-dependent absorption response and the learned residual model introduces experimentally observed perturbations. The analysis is used as a diagnostic control and does not imply that the residual statistics are strictly independent of concentration or identical across all measurement campaigns.

2) RMS-Based Consistency of AE–GMM Noise Generation: The number of Gaussian components in the AE latent space was selected using the Bayesian information criterion, as described in the model-development section. The consistency of the final AE–GMM generator was then assessed by comparing the RMS-amplitude distributions of experimental residuals and generated perturbations under the same representative 45-m configuration.

Fig. 7 shows that the generated perturbations reproduce the main RMS-amplitude distribution of the experimental residuals, including its central region and higher-amplitude range. This agreement supports the use of the generated perturbations in the semi-synthetic training dataset.

The AE–GMM generator reproduces the main amplitude range of the experimental residuals, although this RMS comparison alone cannot describe their complete spectral structure. The key question is therefore whether the generated perturbations are sufficiently representative for training. This is examined in the following sections through CNN prediction on independent experimental measurements.

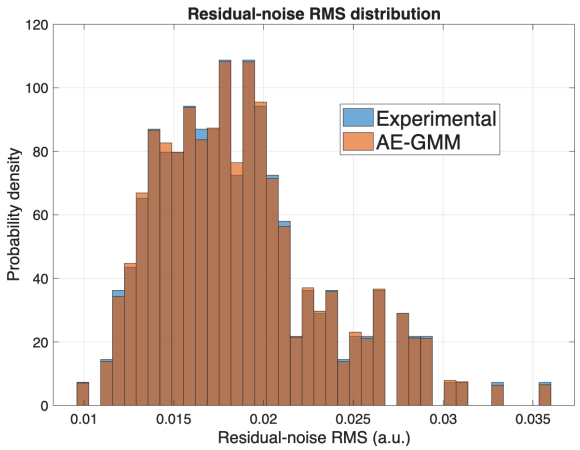


Fig. 7: Comparison of the RMS-amplitude distributions of experimental residuals and AE–GMM-generated perturbations for the representative 45 m measurement configuration.

TABLE III: Global concentration retrieval performance computed on the experimental test dataset.

Metric	Value
Number of spectra, N	886
Mean Absolute Error (MAE)	13.43 ppm
Root Mean Square Error (RMSE)	17.01 ppm
Mean signed error (Bias)	0.30 ppm
Standard deviation of signed error	17.01 ppm

B. Experimental Concentration Retrieval Performance

The CNN was evaluated on experimental spectra acquired independently from the semi-synthetic data used for model construction and training. The purpose of this evaluation was to determine whether the spectral-to-concentration relationship learned from semi-synthetic data could be transferred to real UV–DOAS measurements.

For the comparative analysis, three experimental campaigns at nominal source–detector distances of 15, 45, and 80 m were retained. These campaigns provided sufficient concentration coverage within the experimental conditions considered in the present study.

Fig. 8 compares the predicted and reference concentrations for the three retained experimental campaigns. The dashed line represents the ideal response. The shaded region corresponds to the predefined acceptance limits of ± 37.5 ppm below 250 ppm and $\pm 15\%$ at concentrations of 250 ppm and above. Circular and cross-shaped markers indicate predictions within and outside these limits, respectively.

Overall, the predictions remain closely distributed around the reference concentrations over the investigated experimental range, showing that the relationship learned from semi-synthetic data transfers effectively to real measurements. Prediction dispersion nevertheless varies across concentration levels and experimental campaigns, indicating that the remaining errors are not uniformly distributed across the measurement conditions.

The corresponding global performance metrics are summarized in Table III. They were calculated over the complete experimental test set using the signed error $e = \hat{C} - C_{\mathrm{ref}}$.

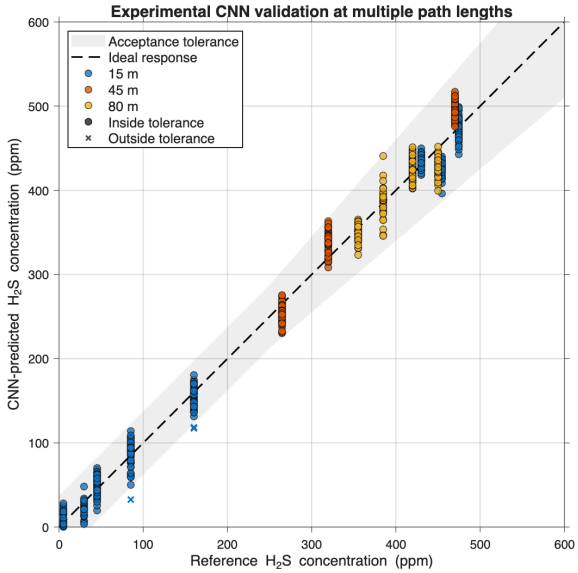


Fig. 8: CNN-predicted versus reference H_2S concentrations for experimental spectra acquired during three measurement campaigns at nominal source–detector path lengths of 15, 45, and 80 m. Colors identify the measurement campaigns. The dashed line represents the ideal response, and the shaded region indicates the predefined acceptance limits. Circular and cross-shaped markers denote predictions within and outside these limits, respectively.

The differences between the individual measurement campaigns are examined in Section VI-C.

C. Performance Across Experimental Campaigns

The concentration retrieval performance was further analyzed across the three experimental campaigns identified by their nominal source–detector path lengths of 15, 45, and 80 m. These measurements do not constitute a controlled distance sweep, since source conditions, signal level, and other acquisition parameters were not varied independently between campaigns. The distance labels therefore identify distinct experimental subsets rather than an isolated propagation-length variable.

TABLE IV: Concentration retrieval performance for the three retained experimental campaigns, identified by their nominal source–detector path length.

Campaign (m)	N	MAE (ppm)	RMSE (ppm)	Bias (ppm)
15	382	12.23	15.78	-3.21
45	181	18.93	21.99	9.69
80	323	11.77	15.10	-0.80

As summarized in Table IV, retrieval performance differs between the three experimental campaigns but does not show a monotonic degradation with increasing nominal path length. In particular, the 80-m campaign remains comparable to the shorter-path measurements, showing that propagation distance alone does not explain the observed prediction dispersion.

These results demonstrate transfer of the same CNN estimator across distinct experimental campaigns and acquisition conditions spanning nominal source–detector separations from 15 to 80 m. Because propagation distance and campaign-specific acquisition parameters were not varied independently, the observed differences cannot be attributed to path length alone.

D. Error Distribution and Concentration-Dependent Performance

The global signed error exhibited a mean bias of 0.30 ppm and a standard deviation of 17.01 ppm. Since the bias remains small relative to the RMSE of 17.01 ppm, the overall retrieval error is primarily associated with dispersion rather than with a uniform systematic offset. These aggregate values nevertheless combine spectra acquired at different concentration levels and under different campaign-specific conditions.

The retrieval performance was therefore further examined as a function of reference concentration to determine whether specific concentration ranges contributed disproportionately to the global error. The concentration intervals were selected to distinguish the low-concentration regime and to account for the change in the acceptance criterion at 250 ppm. The concentration-stratified metrics, calculated from the same experimental dataset used for Fig. 8 and Tables III and IV, are summarized in Table V.

TABLE V: Concentration-dependent retrieval performance computed on the experimental test dataset.

Reference concentration range (ppm)	N	MAE (ppm)	RMSE (ppm)	Bias (ppm)
$0 \leq C_{\mathrm{ref}} < 100$	182	9.93	12.70	1.16
$100 \leq C_{\mathrm{ref}} < 250$	61	14.76	14.76	-7.76
$C_{\mathrm{ref}} \geq 250$	643	14.55	18.23	0.83

This concentration-stratified analysis complements the global and campaign-specific metrics by identifying whether the remaining retrieval error is concentrated within a particular operating range. The results must nevertheless be interpreted jointly with the campaign analysis, since the concentration ranges contain spectra acquired under different experimental configurations.

Overall, the experimental results demonstrate that the semi-synthetic training strategy transfers successfully to real UV–DOAS measurements acquired in three experimental campaigns at nominal source–detector distances of 15, 45, and 80 m. The retained datasets cover several concentration levels and were acquired under the nominal amplification configurations represented in the experimental noise corpus used for training.

Retrieval performance varies between campaigns, but no monotonic degradation is observed with increasing path length. The limited global bias indicates that the remaining error is mainly associated with prediction dispersion rather than with a systematic offset. Since propagation distance, source stability, amplification settings, and other acquisition conditions were not varied independently, the differences observed between the three campaigns cannot be attributed to distance alone.

VII. Discussion and Limitations

The experimental results show that a CNN trained exclusively on semi-synthetic spectra can be transferred to independent UV–DOAS measurements of H_2S acquired across three experimental campaigns at nominal source–detector distances of 15, 45, and 80 m. The semi-synthetic dataset combines a physically constrained absorption model with experimentally learned spectral perturbations. The observed errors remain mainly governed by dispersion rather than by a strong systematic bias, supporting the relevance of the proposed semi-synthetic calibration strategy. The present validation nevertheless defines a specific experimental domain within which the reported performance should be interpreted.

A. Environmental and Thermodynamic Variability

The retained experimental campaigns were acquired under different source–detector separations and acquisition periods and therefore include variations in temperature, humidity, optical attenuation, atmospheric scattering, and instrumental conditions. These variables are not provided explicitly to the regression model.

Environmental variability is instead represented indirectly through the spectral residuals learned by the noise model and through the diversity of the available experimental datasets. The method should therefore be considered validated within the environmental envelope covered by the present measurements. Extrapolation to substantially different atmospheric or thermodynamic conditions would require additional experimental data and, if necessary, explicit environmental inputs.

B. Propagation Distance and Generalization Domain

The source–detector distance is not provided as an input to the CNN. The same regression model is therefore applied unchanged to the three retained experimental campaigns acquired at nominal path lengths of 15, 45, and 80 m, without distance-specific calibration.

The experimental results do not show a monotonic degradation of retrieval performance with increasing path length. In particular, the 80-m measurements remain comparable to those obtained at shorter distances. This indicates that, within the investigated dataset, propagation distance alone does not determine the retrieval error.

The three campaigns, however, do not constitute a controlled distance sweep. Source stability, amplification settings, signal level, and other acquisition conditions were not varied independently of propagation distance. The campaign-to-campaign differences observed in the results therefore reflect the combined influence of propagation and acquisition conditions and cannot be attributed quantitatively to distance alone.

The reported results consequently demonstrate transfer of the same estimator across experimental configurations spanning nominal source–detector separations from 15 to 80 m, but they do not establish a general distance-dependent performance law. A controlled acquisition protocol in which propagation distance and instrumental conditions are varied independently would be required to isolate this effect.

C. Temporal Drift and Instrument Stability

The present framework does not include an explicit temporal model of instrumental drift. Long-term variations such as gradual source aging, optical misalignment, detector response changes, or electronic drift are therefore not parameterized as independent inputs.

Part of this variability may be implicitly included in the experimental residuals used to train the noise model, since the measurements were acquired over extended sessions and across different acquisition periods. Nevertheless, this implicit treatment does not replace a dedicated long-term stability study. For extended deployment, periodic recalibration, drift monitoring, or retraining of the noise model may be required.

D. Scope, Limitations, and Future Work

The present validation is limited to single-gas H_2S measurements within the investigated UV spectral range. The method was not evaluated under multi-gas interference conditions or in the presence of additional absorbing species with overlapping spectral signatures. Its extension to chemically more complex environments therefore remains to be demonstrated.

Preliminary model-development tests also showed that experimental transfer performance is sensitive to the input representation, feature-aggregation strategy, and preprocessing parameters. The reported results should therefore be interpreted as the performance of the complete retained estimation pipeline, rather than as an architecture-independent property of the semi-synthetic dataset. A systematic ablation of these components, together with a controlled experimental protocol designed to isolate propagation distance from instrumental and acquisition conditions, will be addressed in future work.

The proposed approach is best understood as a data-driven measurement framework for UV absorption spectra affected by broadband attenuation, baseline distortions, and experimentally observed perturbations. It is not intended to replace conventional spectroscopic inversion under well-controlled measurement conditions, but to provide a complementary concentration-retrieval strategy for degraded open-path measurements in which low signal levels, baseline variability, and structured experimental noise can reduce the reliability of direct inversion.

Finally, the reported MAE, RMSE, bias, and acceptance-envelope results provide an empirical characterization of retrieval performance under the investigated experimental conditions. They do not constitute a complete metrological uncertainty budget within the GUM framework [22]. Establishing such a budget would require a dedicated evaluation of the individual uncertainty contributions, including reference-concentration uncertainty, wavelength and intensity calibration, optical alignment, source stability, environmental variability, and preprocessing-induced effects.

VIII. Conclusion

This work presents a semi-synthetic training framework for ppm-level H_2S concentration retrieval via UV-DOAS spectroscopy. The approach combines a Beer–Lambert forward model with structured noise learned directly from long-term

outdoor acquisitions, without relying on any analytical noise parameterization. All performance evaluations are carried out on purely experimental measurements, with no overlap between training and test data.

The results consistently show that noise representation fidelity matters more than regression architecture. The proposed semi-synthetic framework achieves MAE = 22.8 ppm across the full experimental envelope, against 45 ppm for the noise-free baseline and 35 ppm for conventional Gaussian augmentation — without distance-specific calibration, explicit spectral inversion, or temporal modeling.

Validation over several years of outdoor acquisitions spanning 15 m to 80 m under diverse atmospheric conditions confirms that the framework holds under realistic field constraints. The MAE increases with distance, as expected from propagation-induced perturbations not fully represented in the training corpus at extended ranges, but remains within levels reported for comparable open-path monitoring systems.

The present validation covers controlled H_2S/N_2 mixtures over the explored distance and atmospheric envelope. Conditions beyond this range — extreme humidity, dense aerosol loading, or propagation distances significantly exceeding 100 m — require dedicated noise retraining and remain open for future investigation. The methodology extends naturally to other UV-absorbing species such as NH_3, given appropriate reference cross-sections and experimental noise statistics; multi-gas retrieval is beyond the scope of the present work.

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